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Database Systems (CSF212)

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Triggers

- Enable the database programmer to specify:
- when to check a constraint,
- what exactly to do.

- A trigger has 3 parts:
  - An event (e.g., update to an attribute)
  - A condition (e.g., a query to check)
  - An action (deletion, update, insertion)

- When the event happens, the system will check the constraint, and
- if satisfied, will perform the action.
- NOTE: triggers may cause cascading effects.
- Triggers not part of SQL2 but included in SQL3… however,
- database vendors did not wait for standards with triggers!

Elements of Triggers (in SQL3)

- Timing of action execution
  - before
  - after
  - instead of
    - …. the triggering event
- The action can refer to both the old and new state of the database.
- Update events may specify a particular column or set of columns.
- A condition is specified with a WHEN clause.
- The action can be performed either for
  - once for every tuple, or
  - once for all the tuples that are changed by the database operation.

Example: Row Level Trigger

```
CREATE TRIGGER    NoLowerPrices

AFTER UPDATE OF  price  ON Product //price is an attribute and
Product is the table
REFERENCING
    OLD  AS OldTuple
    NEW  AS  NewTuple
WHEN (OldTuple.price > NewTuple.price)
    UPDATE  Product
    SET  price = OldTuple.price
    WHERE  name = NewTuple.name

FOR EACH ROW
```

Statement Level Trigger

emp(dno…), dept(dept#, …)

"Whenever we insert employees tuples, make sure that their dno's exist in Dept."

```
CREATE TRIGGER  deptExistTrig
AFTER INSERT ON emp
REFERENCING
    OLD_TABLE  AS OldStuff
    NEW_TABLE AS  NewStuff
WHEN (EXISTS (SELECT * FROM NewStuff
                     WHERE dno NOT IN
                     (SELECT dept# FROM dept)))
DELETE FROM NewStuff
             WHERE dno NOT IN
                (SELECT dept# FROM dept));
```